

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R56 responds to the 9 October 2026 advisory report on R54. The same Neural Bellman Operators paper retains its original constructive and economic targets and adds trained-neural action witnesses, non-tensor verification, prospective stopping and confidence-qualified learned structure.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators — Revision R56*

We thank the referee for the detailed assessment of the R54 manuscript and for distinguishing its valid directed policy account from the stronger computational and economic claims still requiring evidence. The present revision develops the same **Neural Bellman Operators** paper. It retains the original title, economic laws, own-future construction, feasible original-owner compiler, primitive resource account, Bellman-optimality target and economic implementation-price results. It adds a trained-neural action-search theorem, a non-tensor cached implementation, signed smooth-reference sensitivity, prospective stopping and confidence-qualified learned structure.

The controlling advisory report is dated 9 October 2026 on review commit adf1256cff9cde365246a3db2dac90c72fda3b13, reviewing manuscript 00e837adb431f4d4b5248fc6d5ff1fc927dd3b65. It is an owner-commissioned advisory report, not a decision of the Econometric Society. The repository subsequently contains separately frozen R55 primary and signed-sensitivity executions, culminating at 7f4de488134e93b6cfb5dd95a3e6d7e14b45ac2a. Their main-paper wrappers had not yet incorporated those results. R56 integrates that evidence, supplies complete mathematical statements and proofs, and independently checks saved numerical accounts without adding cost samples or rewriting service times.

The active manuscript is reorganized around the original economic operator and constructive resource result, trained neural continuations, whole-cell policy verification, prospective economic return and actual policy costs. Its technical supplement contains the entire service registry. The preceding active article and supplement are retained as development documents, and both complete development companions retain the full earlier body. This is not the alternative retitled paper suggested in the report: the original NBO subject and substantive results remain intact.

1.2. *B1, M1: A specific trained-neural contribution*

The new neural continuation is genuinely fitted: all hidden and output coefficients receive Adam updates before a final ridge output fit, and its stored dyadic coefficients are retained. It is not a relabeled native min-plus envelope. The integrated-continuation lemma gives the exact scalar-uniform expectation of each learned ridge feature, with outward numerical evaluation and explicit treatment of zero innovation projection.

The new **Algebraic neural action witnesses** theorem then reduces the original scalar investment objective to piecewise quartics. Rational hinge breakpoints and cubic stationary-root isolation yield an exact minimizer on the deployed action lattice using at most $22m+13$ candidate indices for m hidden features, with coefficient bit lengths and root-isolation cost separately charged. This supplies a terminating fitted-critic action search rather than assuming another numerical optimizer succeeds. The code uses exact rational comparisons and handles ties, negative output weights and zero projected noise.

The solver agrees with exhaustive rational lattice minimization on the retained saved-neural fixtures in dimensions two, four and eight. These are transparent solver regressions, not new actual-cost observations. The R55 production design used its frozen 17-action construction menu; we have not reassigned its economic intervals or clocks to policies generated by the new search. The original constructive-backend theorem and compiler remain in the active paper as a distinct route to primitive Bellman accuracy.

1.3. *B2, B6, M6: What actually determines the improved policy?*

The comparison includes trained ReLU, quadratic FVI and ExtraTrees FVI in all three dimensions, with the original compiled-witness and tensor-FVI frontends in the two-dimensional task. All methods receive the same partition, common robust candidate actions, continuous-action lower cover, reference information and economic stopping rule. None receives a free policy-value oracle or free inference.

We add a complete deterministic candidate-attribution replay. Removing the tenth, generator-specific proposal does not change any deployed neural, quadratic or tree action in these cohorts. The only proposal-specific effect is one compiled-witness cell in the two-dimensional sensitivity experiment, repeated across the two targets and three timing runs. In dimensions four and eight the sensitivity policies are exactly identical across generators. These identities are checked by complete policy arrays, not inferred from overlapping intervals.

The revised paper therefore assigns the observed improvement to the whole implemented procedure, while identifying which parts are genuinely trained-neural computations. At the ten-percent target, quadratic FVI has the smallest observed median whole-process time in each task. The earlier unfavorable witness comparisons and the stronger conventional comparators remain in the development edition. The paper neither deletes those outcomes nor calls an unresolved or identical-policy comparison a neural victory.

1.4. *B3, M3, M4: Accuracy, enclosure width and the dominant obstruction*

The full-action directed theorem is retained with its statewise safety statement, coefficient-one error recursion, finite-horizon expansion and sharpness example. The new cache proposition makes its required intervals computable on a non-tensor partition. It encloses residuals and values on each assigned observation cell, retains every future leaf intersecting a query box, and intersects independently valid residual and direct-value accounts. Discontinuous actors are not assigned a fictitious derivative.

The primary cache-only cohort exposes the remaining obstruction: broad continuation and actor ranges block nonterminal changes. The separately frozen signed-

sensitivity extension addresses that obstruction for the installed smooth zero reference. Its complete proof propagates state boxes forward, sparse derivative enclosures backward and signed action projections along the first-action segment. It never applies this derivative argument to an acquired policy after that policy has changed.

The eight-dimensional neural example reduces the maximum date-zero candidate width from more than 14 to less than 0.610 and changes nonterminal cells at five dates. The paper gives every date's U-C, C-L and U-L account, with the distinction between a maximum of sums and a sum of maxima. It also gives a monotone refinement ledger for independent valid enclosure refinements. These measured accounts do not purport to be a separately executed causal decomposition into six numerical error sources.

The original all-state Bellman target is not replaced by the new expected-cost target. The remaining all-state bound is reported and remains conservative; one successful prospective sweep is not called exact Bellman convergence. Both the full-action accuracy theorem and the actual initial-law economic criterion are explicit.

1.5. *B4, M2: A prospective from-primitives stopping service*

Each method/task/target combination starts from primitives in its own process. The two resource stages, acquired leaves, training budgets, inference looks and expected-cost targets are specified before that cohort's production. At each stage the complete new policy is frozen before its fresh inference stream. The service returns at the first nonpositive upper bound for $J(\text{policy}) - q \backslash J(\text{installed})$. A strictly positive lower bound advances the resource stage; an unresolved interval advances the declared look. No later stage or look is computed after a successful return.

The first-attainment theorem proves the economic validity of this return under a finite conditional confidence allocation. It also proves that every returned incumbent, including a budget-exhausted one, retains deterministic statewise safety. All own construction, verification, inference, serialization and durable output are charged by the surrounding process clock. This is not the R54 retrospective complete-catalogue frontier relabeled as a prospective algorithm.

The full 66-process primary cohort has 39 target attainments and 27 budget exhaustions. The separately frozen sensitivity cohort has 66 target attainments, all at its first resource stage. Its design was motivated by the completed primary result and is identified as such. The initial controller’s interrupted attempt and subsequent immutable-checkpoint correction are retained; no failed execution is silently removed or assigned zero work.

1.6. *B5, M5: Non-tensor execution and explicit dimensional work*

The new acquired-state representation is a locally split binary partition with 32 or 128 leaves, rather than a Cartesian product of coordinate grids. The complete services run the original nonlinear family at $(d, T) = (2, 2), (4, 4), (8, 6)$. This is a genuinely multidimensional non-tensor implementation, not a nonuniform tensor grid described as non-tensor.

The work proposition charges the literal cell-membership calculation as $O(T A q N^2 d)$ plus primitive and critic work, where N is the number of leaves and A counts candidates and covering intervals. The signed smooth-reference calculation costs $O(A N q d T^2)$ and can stream its innovation children. It needs neither a full state-pair table nor enumeration of q^T future innovation histories. The primitive Jacobian gives the new dimension-uniform smooth-reference stability bound 1920/91 for the original normalized family.

This bound and the executed eight-dimensional service are positive, specific results. They do not imply that a fixed leaf count resolves arbitrary dimensional models at a fixed all-state accuracy. The manuscript retains that separate approximation burden in its explicit work and error accounts rather than presenting non-tensor storage as a universal elimination of dimensional complexity.

1.7. *B7, M7: Learned directions and uncertain action exposure*

The new sample-split experiment fits directions from observations instead of selecting an exact invariant by hand. Independent validation data give a coordinate confidence box for an unknown action-exposure vector in $[0, 1]^d$. Both the

nuisance correction and the remaining critic advantage are evaluated against the same box; the corrected gate does not receive the true simulator parameter.

The confidence-qualified correction theorem holds uniformly over fitted directions, amplitudes and tested actions on that box event. The full record contains nine fitted directions, 27 validation boxes, 81 amplitude cases and 5,184 cell-case decisions. All recorded boxes cover the true exposure, and the corrected gate makes 165 safe changes. No strictly harmful plug-in choice is identified in this catalogue. That zero is reported, not replaced by the much larger harmful count from the earlier, different known-perturbation exercise.

The maintained model has a known nonlinear drift and bounded additive action-invariant noise; the inferred object is its action exposure. The theorem does not presume that fitting a direction identifies an arbitrary unknown kernel. The primary cost, sensitivity cost and learned-exposure families have a joint union error at most 1/40, without treating reused policies or timing repetitions as new independent observations.

1.8. B8, M8: Repetitions, hardware and statistical units

Every declared service has three isolated sequential process runs, a fixed available core and one numerical-library thread. Method order rotates across repetitions. The tables report the minimum, median and maximum complete-process times, output bytes and peak resident memory. All own inference and durable records are included.

The repetitions deliberately keep training and inference seeds identical. Exact policy and confidence-interval identities are verified, so they are timing/reproducibility repetitions, not a sample of independently trained critics. Clock frequency is not controlled, and comparisons are within a task runner rather than a cross-runner hardware-scaling regression. The new independently fitted direction observations belong to the separate structural-uncertainty study and are not counted as independent neural-policy fits. No optimizer success probability is inferred from these records.

1.9. B9, M9: *The actual economic return*

The original investment costs and transition laws are unchanged. The targets require two- and ten-percent expected-cost reductions from a specified installed zero-investment controller. In the eight-dimensional, six-date neural service, changed leaf counts are (26,24,24,20,14,0): the successful policy changes are nonterminal, not a relabeled final-date repair. At the ten-percent return, the cumulative gain interval is $[0.21452658, 0.29774911]$ with outward-rounded endpoints.

The adoption corollary maps each gain interval into the entire replacement-fee and computation-price plane: net gain lies in $[l\text{-}\tau\text{-}\lambda W, u\text{-}\tau\text{-}\lambda W]$. It gives both the supported adoption region and the region where adoption is ruled out, with the intermediate region left unidentified. This is an actual implemented-policy decision with paid computation, not a conclusion about an unchanged policy's certificate. The quantities remain normalized model units rather than a claimed external calibration.

1.10. B10, M10: *A coherent original-paper exposition without deletion*

The main article now follows one connected path: economic operator, constructive original-owner and resource theorems, trained neural continuations and exact action witnesses, non-tensor directed verification, economic inference and prospective return. The theoretical expected-cost scope is stated before the new linear-kernel and sampling results. The broader recursive-utility, diffusion, temporal-self and game results retain their own assumptions in the complete companions.

The previous active main and supplement are also rebuilt as development documents. All inherited theorem labels and immutable section/source/evidence hashes are checked in the companions. The original resource-allocation result remains in the main article, with its historical empirical context clearly identified. Earlier adverse results are preserved, not edited to make the new comparison look favorable. This editorial consolidation preserves the original paper's substantive development while making its current theorem–implementation–economic chain readable.

1.11. *Verification supplied with the revision*

The new audit reloads every saved fitted model and regenerates all 33 unique primary and sensitivity certificates. It compares every stored certificate endpoint and deployed action, replays all 132 services and 354 stopping looks, verifies exact incumbent transfer and finite-sweep recurrences, and checks all learned-direction decisions. Original sample means, second moments and variances are verified by exact rational arithmetic before their stored confidence endpoints are reconstructed. This avoids mistaking cross-platform differences in final floating-point reductions for new economic observations.

The inherited tests and new algebraic-search, stability, error-account and stopping regressions are included. Publication builds the main article, technical supplement, response and four complete/historical companions. A clean archive independently regenerates publication tables and documents after checking the full scientific-source/result binding; it is explicitly a clean document reproduction, not a second execution of the scientific services. Final delivery is conditioned on the release manifest and confirmed remote branch identities. These verification operations are evidence of reproducibility and internal consistency, not a claim of external mathematical or editorial acceptance.